of this toxin is scalded skin syndrome. The patient develops painful rash which slough off and skin surface resembles scalding.

HOTS

Why most infections acquired through the skin are non-communicable diseases?

Toxic shock syndrome toxin: It is caused by TSST-1 and characterized by high fever, hypotension (low blood pressure), vomiting, diarrhea and erythematous rash. TSS became widely known in association with the use of vaginal tampons by menstruating women but it occurs in other situations also.

7.3.5 Laboratory Diagnosis

Specimens: The clinical specimens are collected according to the nature of Staphylococcal infections, which is given in the (Table 7.2).

Table 7.2: Clinical specimen collected for Staphylococcal infections

Infections	Clinical Specimens
Suppurative lesions	Pus
Respiratory infections	Sputum
Septicemia	Blood
Meningitis	CSF
Food poisoning	Faeces, food or vomitus

Specimens should be transported immediately to the laboratory and processed.

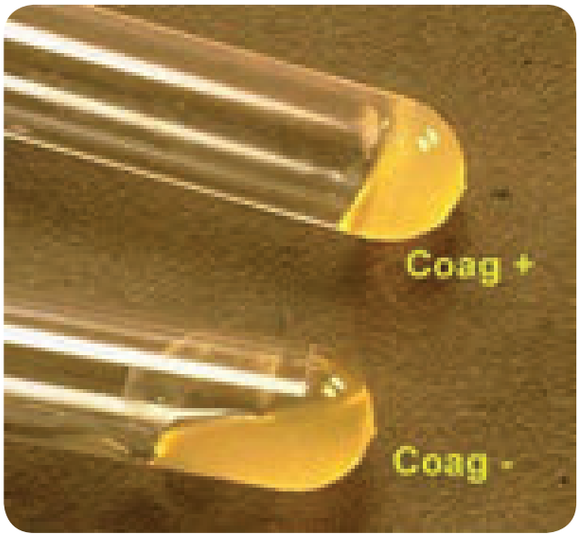


Figure 7.3: Skin infections caused by *Staphylococcus aureus*

Direct Microscopy: Gram stained smears of clinical specimens is done, where gram positive cocci in clusters were observed.

Culture: The collected specimen is inoculated on selective media-MSA and the media incubated at 37°C for 18–24 hours. Next day culture plates are examined for bacterial colonies, which are identified by gram staining, colony morphology and biochemical tests such as

a. Catalase test: The genus *Staphylococci* are catalase positive. This test distinguishes *Staphylococcus* from *Streptococcus* (catalase negative).

b. Coagulase test: This test helps in differentiating a pathogenic strain from non-pathogenic strain. *S. aureus* is coagulasepositive (Figure 7.3).

7.3.6 Treatment

Benzyl penicillin is the most effective antibiotic. Cloxacillin is used against beta lactamase. Producing strains (β-lactamase

TN_GOVT_XII_Micro_Biology_CH07.indd 118 15-04-2019 16:38:24